

Boss Challenge — Paper 39 (Class 7)

Light | Electric Current & its Effects | Symmetry | Perimeter & Area

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. In clear air light travels along paths that are perfectly:
(A) zig-zag steps
(B) spreading circles
(C) straight lines
(D) curved arcs
2. For a bulb to glow, the electric circuit joining it to the cell must form a:
(A) single straight wire
(B) broken open path
(C) closed (complete) loop
(D) loop with the switch off
3. A figure has line symmetry if a straight line divides it into two halves that are exact:
(A) mirror images of each other
(B) rotated by 90°
(C) different shapes
(D) bigger and smaller copies
4. The perimeter of any flat figure is the total length of its:
(A) boundary all the way around
(B) inside region
(C) two diagonals
(D) longest side only
5. We are able to see an ordinary book only because light:
(A) passes straight through it
(B) is made inside our eyes
(C) is stored in the book
(D) from it enters our eyes
6. Inside any household circuit, the part whose job is to open or close the path for current is the:
(A) cell
(B) bulb
(C) fuse
(D) switch
7. The straight line along which a symmetric figure can be folded so the halves match is called the:
(A) line of symmetry
(B) base line
(C) diagonal only
(D) number line

8. The perimeter of a rectangle of length l and breadth b is:
- (A) $4 \times l$
 - (B) $2 \times (l + b)$
 - (C) $l + b$
 - (D) $l \times b$
9. A shadow forms behind an object only when that object is:
- (A) transparent
 - (B) glowing brightly
 - (C) opaque to light
 - (D) painted white
10. A glowing torch bulb gives off light because its thin filament gets so hot it shines. This is the:
- (A) magnetic effect of current
 - (B) cooling effect of current
 - (C) heating effect of current
 - (D) chemical effect of current
11. A square has exactly this many lines of symmetry:
- (A) 2
 - (B) 1
 - (C) 8
 - (D) 4
12. If every side of a square measures s , then the total distance around its boundary works out to:
- (A) $s + 4$
 - (B) $4 \times s$
 - (C) $2 \times s$
 - (D) $s \times s$
13. When a ray of light strikes a plane mirror, the angle it bounces off at equals the angle it:
- (A) doubles to
 - (B) halves to
 - (C) arrived at
 - (D) turns 90° from
14. The filament of a bulb is made of tungsten because tungsten has a very high:
- (A) colour
 - (B) cost
 - (C) weight
 - (D) melting point
15. A rectangle that is not a square has exactly this many lines of symmetry:
- (A) 0
 - (B) 4
 - (C) 1
 - (D) 2

16. To find how much surface a rectangle of length l and breadth b covers, you compute:

- (A) $l + b$
- (B) $2 \times (l + b)$
- (C) $l \times b$
- (D) $4 \times l$

17. The image of your face in a flat bathroom mirror is:

- (A) upside down
- (B) the same size as your face
- (C) larger than your face
- (D) smaller than your face

18. A safety fuse protects an appliance by melting and breaking the circuit when the current becomes:

- (A) too small
- (B) too large
- (C) perfectly steady
- (D) turned off

19. An equilateral triangle (all sides equal) has this many lines of symmetry:

- (A) 0
- (B) 1
- (C) 3
- (D) 2

20. A square sheet of card measures 6 cm along each edge; the surface it covers is:

- (A) 36 cm^2
- (B) 24 cm^2
- (C) 12 cm^2
- (D) 36 cm

21. Hold up your right hand to a flat mirror and the image seems to raise its left hand. This left-right swap is called:

- (A) magnification
- (B) lateral inversion
- (C) refraction
- (D) dispersion

22. Hans Oersted discovered that a wire carrying current can deflect a nearby compass needle, showing that current has a:

- (A) heating effect
- (B) magnetic effect
- (C) chemical effect
- (D) lighting effect

23. A scalene triangle, whose three sides are all different lengths, has this many lines of symmetry:
- (A) 2
 - (B) 1
 - (C) 0
 - (D) 3
24. The area of a triangle with base b and height h is:
- (A) $b \times h$
 - (B) $b + h$
 - (C) $2 \times b \times h$
 - (D) $\frac{1}{2} \times b \times h$
25. An image formed by a plane mirror cannot be caught on a screen, so it is called a:
- (A) virtual image
 - (B) inverted image
 - (C) real image
 - (D) solid image
26. A coil of wire wound on an iron piece becomes a magnet only while current flows. This device is an:
- (A) simple battery
 - (B) electromagnet
 - (C) ordinary fuse
 - (D) permanent magnet
27. A circle has this many lines of symmetry:
- (A) only 2
 - (B) 0
 - (C) only 4
 - (D) infinitely many
28. The area of a parallelogram is found by multiplying its base by its:
- (A) perimeter
 - (B) height (perpendicular distance)
 - (C) two diagonals
 - (D) slanted side
29. A mirror whose reflecting surface curves inward, like the inside of a spoon's bowl, is a:
- (A) plane mirror
 - (B) plain glass
 - (C) concave mirror
 - (D) convex mirror

30. An electric bell rings repeatedly because it uses an electromagnet that switches:
- (A) to a different colour
 - (B) on and stays on
 - (C) on and off rapidly
 - (D) off permanently
31. The capital letter H has this many lines of symmetry:
- (A) 2
 - (B) 4
 - (C) 1
 - (D) 0
32. The distance once around a circle is called its:
- (A) area
 - (B) circumference
 - (C) diameter
 - (D) radius
33. A convex (outward-bulging) mirror always gives an image that is upright and:
- (A) exactly equal in size
 - (B) upside down
 - (C) larger than the object
 - (D) smaller than the object
34. Materials such as copper and iron let current pass through them easily, so they are called:
- (A) insulators
 - (B) conductors
 - (C) magnets
 - (D) fuses
35. When a figure looks exactly the same after being turned about its centre by less than a full turn, it has:
- (A) translation symmetry
 - (B) line symmetry only
 - (C) no symmetry
 - (D) rotational symmetry
36. The circumference of a circle of radius r is given by:
- (A) $\pi \times r$
 - (B) $2 \times \pi \times r$
 - (C) $2 \times r$
 - (D) $\pi \times r^2$

37. When white sunlight passes through a glass prism it spreads into a band of seven colours. This splitting is called:
- (A) lateral inversion
 - (B) reflection
 - (C) absorption
 - (D) dispersion
38. Rubber and plastic do not allow current to pass, so wires are coated with them to act as:
- (A) insulators
 - (B) electromagnets
 - (C) cells
 - (D) conductors
39. Turning a square about its centre through one full revolution, the number of times it lands exactly on itself is:
- (A) 1
 - (B) 8
 - (C) 2
 - (D) 4
40. The area of a circle of radius r is:
- (A) $\pi \times d$
 - (B) $\pi \times r^2$
 - (C) $2 \times \pi \times r$
 - (D) $\pi \times r$
41. The seven colours of the rainbow, in order from top to bottom, begin with violet and end with:
- (A) red
 - (B) green
 - (C) white
 - (D) blue
42. If you connect two identical cells in series instead of one, the bulb in the same circuit usually glows:
- (A) exactly the same
 - (B) more dimly
 - (C) a different colour
 - (D) more brightly
43. The smallest angle through which an equilateral triangle must be turned to look the same is:
- (A) 90°
 - (B) 60°
 - (C) 120°
 - (D) 180°

44. For school problems, the value of π is most often taken as:
- (A) $7/22$
 - (B) $2/7$
 - (C) $22/7$
 - (D) 22
45. A lens that is thicker in the middle than at its edges bends parallel rays to meet at a point, so it is called a:
- (A) converging lens
 - (B) diverging lens
 - (C) opaque disc
 - (D) plane mirror
46. A battery, strictly speaking, is made of:
- (A) two or more cells joined together
 - (B) a bulb and a wire
 - (C) a switch and a fuse
 - (D) a single cell only
47. A regular polygon with n equal sides always has this many lines of symmetry:
- (A) 2 always
 - (B) n
 - (C) $n - 1$
 - (D) always 4
48. The perimeter of a rectangular field is 30 m and its length is 9 m. Its breadth is:
- (A) 3 m
 - (B) 21 m
 - (C) 12 m
 - (D) 6 m
49. The Sun is a luminous object, but the Moon is non-luminous because it only:
- (A) reflects sunlight to us
 - (B) blocks all sunlight
 - (C) absorbs all the light
 - (D) makes its own light
50. In a circuit diagram, the symbol for a cell shows a longer line and a shorter line, where the longer line marks the:
- (A) positive terminal
 - (B) bulb
 - (C) negative terminal
 - (D) switch

51. A regular hexagon (6 equal sides) has this many lines of symmetry:
- (A) 6
 - (B) 3
 - (C) 2
 - (D) 12
52. Two shapes can have the SAME perimeter yet different areas. This shows that perimeter and area:
- (A) cannot be compared
 - (B) measure different things
 - (C) are always equal
 - (D) are the same measurement
53. A real image, unlike a virtual one, has the special property that it can be:
- (A) formed only by flat mirrors
 - (B) seen only in a mirror
 - (C) caught on a screen
 - (D) made larger forever
54. An electromagnet can be made stronger by increasing the number of:
- (A) switches added
 - (B) turns in its coil
 - (C) bulbs in the circuit
 - (D) iron cores removed
55. A plain parallelogram (not a rectangle or rhombus) has 0 lines of symmetry but its order of rotational symmetry is:
- (A) 2
 - (B) 0
 - (C) 1
 - (D) 4
56. One square metre (1 m^2) equals this many square centimetres:
- (A) $1\,000 \text{ cm}^2$
 - (B) 100 cm^2
 - (C) $10\,000 \text{ cm}^2$
 - (D) $1\,000\,000 \text{ cm}^2$
57. A pinhole camera forms an image of a candle that is real and:
- (A) the same as in a mirror
 - (B) upside down
 - (C) right side up
 - (D) a band of colours

- 58.** An electric room heater and a clothes iron both work mainly on the:
- (A) reflecting effect of current
 - (B) chemical effect of current
 - (C) heating effect of current
 - (D) magnetic effect of current
- 59.** A line of symmetry behaves exactly like a mirror because each half is the reflection of the other. This links symmetry directly to:
- (A) the boiling of water
 - (B) the heating effect of current
 - (C) the mirror image idea from Light
 - (D) the area of a circle
- 60.** The area of a right-angled triangle whose two perpendicular sides are 8 cm and 5 cm is:
- (A) 40 cm²
 - (B) 13 cm²
 - (C) 20 cm²
 - (D) 20 cm
- 61.** A driver's rear-view mirror is convex because, compared with a flat mirror, it gives a:
- (A) wider field of view
 - (B) upside-down view
 - (C) magnified close-up
 - (D) coloured view
- 62.** If a bulb in a working circuit suddenly stops glowing, the most likely simple cause is a:
- (A) the wire being too short
 - (B) break or loose connection in the loop
 - (C) too-bright Sun outside
 - (D) colour of the casing
- 63.** The mirror image of the digit-display word made by a square's four matching positions shows the square also has rotational symmetry of order:
- (A) 4
 - (B) 3
 - (C) 1
 - (D) 2
- 64.** The diameter of a circle whose radius is 7 cm is:
- (A) 3.5 cm
 - (B) 7 cm
 - (C) 49 cm
 - (D) 14 cm

65. In a flat mirror, if you stand 30 cm in front of the glass, your image appears to be:
- (A) 30 cm behind the glass
 - (B) right on the glass surface
 - (C) 60 cm behind the glass
 - (D) 15 cm behind the glass
66. Compared with an old filament bulb, a CFL or LED is preferred at home because it:
- (A) needs no circuit
 - (B) makes more heat
 - (C) uses less electricity for the same light
 - (D) uses more electricity
67. Which capital letter has a horizontal line of symmetry but NOT a vertical one?
- (A) A
 - (B) H
 - (C) T
 - (D) E
68. A square garden has area 81 m^2 . The length of each side is:
- (A) 81 m
 - (B) 40.5 m
 - (C) 18 m
 - (D) 9 m
69. A dentist uses a concave mirror to look at a tooth up close because, held near, it gives an image that is:
- (A) enlarged and upright
 - (B) enlarged but upside down
 - (C) tiny and upright
 - (D) the exact same size
70. A compass needle brought near a straight wire carrying current will:
- (A) deflect from its north-south rest position
 - (B) turn into a battery
 - (C) stay exactly still
 - (D) melt instantly
71. The number of lines of symmetry of a rhombus (a slanted diamond with 4 equal sides) is:
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 4

72. The circumference of a circular flower bed of radius 7 m, taking $\pi = 22/7$, is:
- (A) 154 m
 - (B) 22 m
 - (C) 14 m
 - (D) 44 m
73. When light bounces off a rough, uneven wall it scatters in many directions. This is called:
- (A) refraction
 - (B) regular reflection
 - (C) dispersion
 - (D) diffuse reflection
74. In the symbolic circuit diagram, a switch is shown as a:
- (A) circle with a cross
 - (B) pair of long and short lines
 - (C) zig-zag line
 - (D) small gap that a lever can close
75. An isosceles triangle (exactly two equal sides) has this many lines of symmetry:
- (A) 2
 - (B) 0
 - (C) 1
 - (D) 3
76. A circuit's wire bends right around a rectangular board 12 cm by 8 cm before returning to the cell. The length of wire forming that closed loop equals the:
- (A) diagonal, 20 cm
 - (B) perimeter, 40 cm
 - (C) half-perimeter, 20 cm
 - (D) area, 96 cm
77. A lens that is thinner in the middle than at the edges spreads parallel rays apart, so it is a:
- (A) convex lens
 - (B) plane mirror
 - (C) concave (diverging) lens
 - (D) magnifying glass
78. An MCB (miniature circuit breaker) is used in modern homes in place of a fuse because, after it trips, it can be:
- (A) filled with water
 - (B) eaten away and discarded
 - (C) switched back on without replacing anything
 - (D) left tripped forever

79. A figure whose order of rotational symmetry is 1 actually means the figure:
- (A) matches itself every 90°
 - (B) looks the same only after a full 360° turn
 - (C) cannot be drawn
 - (D) has four lines of symmetry
80. A square plane mirror has a side of 15 cm. The reflecting glass surface you can see covers an area of:
- (A) 60 cm^2
 - (B) 225 cm^2
 - (C) 225 cm
 - (D) 30 cm^2
81. The word AMBULANCE is painted reversed on the front of the van so that a driver ahead reads it correctly in a:
- (A) prism
 - (B) concave mirror
 - (C) plane mirror
 - (D) convex lens
82. The chief job of the iron core inside an electromagnet's coil is to:
- (A) block the current
 - (B) make the magnetism much stronger
 - (C) change the bulb's colour
 - (D) cool the wire down
83. The English capital letter with the GREATEST number of lines of symmetry among A, H, O, T is:
- (A) A
 - (B) T
 - (C) H
 - (D) O
84. To fence a square plot of side 25 m, the length of fencing needed is the plot's:
- (A) side, 25 m
 - (B) area, 625 m
 - (C) diagonal, 50 m
 - (D) perimeter, 100 m
85. Two plane mirrors set face-to-face show many repeated images because the light is:
- (A) bent into the mirror
 - (B) reflected again and again between them
 - (C) absorbed by the glass
 - (D) split into colours

86. When you join cells to make a battery, you must connect the positive terminal of one cell to the _____ of the next.
- (A) negative terminal
 - (B) middle of the wire
 - (C) switch
 - (D) positive terminal
87. A semicircle (half a disc) has this many lines of symmetry:
- (A) 1
 - (B) infinitely many
 - (C) 2
 - (D) 0
88. A rectangle has area 48 cm^2 and length 8 cm. Its breadth is:
- (A) 12 cm
 - (B) 40 cm
 - (C) 6 cm
 - (D) 384 cm
89. A magnifying glass that makes tiny print look bigger is simply a:
- (A) plane mirror
 - (B) convex mirror
 - (C) concave lens
 - (D) convex lens
90. A torch will NOT light if its cell is put in the wrong way round, because the circuit then has its cells:
- (A) joined too tightly
 - (B) made of plastic
 - (C) too cold to work
 - (D) pushing against each other
91. A regular pentagon (5 equal sides) has its order of rotational symmetry equal to:
- (A) 5
 - (B) 4
 - (C) 1
 - (D) 10
92. The area of a circular coin of radius 7 mm, taking $\pi = 22/7$, is:
- (A) 154 mm^2
 - (B) 22 mm^2
 - (C) 49 mm^2
 - (D) 44 mm^2

- 93.** Light from the bright Sun reaches us because, between Sun and Earth, light needs:
- (A) a solid path
 - (B) air all the way
 - (C) a stream of water
 - (D) no material medium to travel through
- 94.** Touching a bare live wire is dangerous because the human body acts as a:
- (A) perfect insulator
 - (B) kind of fuse
 - (C) source of new current
 - (D) conductor of electric current
- 95.** Among these shapes - scalene triangle, square, rhombus, rectangle - the one with the MOST lines of symmetry is the:
- (A) rhombus
 - (B) rectangle
 - (C) square
 - (D) scalene triangle
- 96.** A photo 10 cm by 6 cm is given a uniform border 1 cm wide all around. The outer rectangle (photo + border) measures:
- (A) 11 cm by 7 cm
 - (B) 12 cm by 8 cm
 - (C) 14 cm by 10 cm
 - (D) 10 cm by 6 cm
- 97.** In a periscope used to see over a wall, the light is turned twice using two:
- (A) concave mirrors
 - (B) plane mirrors
 - (C) convex lenses
 - (D) prisms that split colours
- 98.** A drawing that uses agreed symbols for the cell, switch, and bulb instead of real pictures is called a:
- (A) circuit diagram
 - (B) photograph
 - (C) map of a city
 - (D) bar graph
- 99.** The capital letter B has a single line of symmetry, and that line is:
- (A) diagonal
 - (B) horizontal
 - (C) vertical
 - (D) both horizontal and vertical

100. The number of square tiles, each of side 1 m, needed to cover a floor 4 m long and 3 m wide is:

- (A) 12
- (B) 7
- (C) 24
- (D) 14